ShadowWeave: A Calibrated World Model of Occluded Space from a Single Depth Frame

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Abstract

An embodied agent must model the physical scene it cannot directly see — the occupancy occluded by the obstacles it moves among ("shadow"); a world model of only the visible surface is blind to half its workspace. We study how well a model can **complete** occupancy there from a **single monocular-depth frame**, and whether it knows *when it is guessing*. ShadowWeave projects one depth frame into an egocentric bird's-eye-view (BEV) occupancy grid, marks unobserved cells with an explicit shadow mask, and predicts occupancy and per-cell uncertainty inside that mask. On a MuJoCo indoor benchmark, the model beats a persistence baseline by a micro-averaged shadow-IoU gain of **+0.31 to +0.36 at a 5 s horizon** (95% per-episode bootstrap CIs clear of zero at every horizon), far exceeds an observation-blind dataset prior, and its uncertainty is **better calibrated inside shadow (ECE 0.043–0.047, at 5 s) than in directly observed space (0.058–0.086)**. Framing this as world modeling of the unobserved, a decomposition separating *amodal completion* from *temporal forecasting* shows the gain is almost entirely completion — the pure-forecasting residual is small ($\leq$+0.007 at 5 s). Randomizing room geometry per episode leaves the gain statistically unchanged on never-seen rooms, ruling out memorization of a fixed shell. The completion path runs at 7 ms (p95); end-to-end with an off-the-shelf monocular-depth network it is $\sim$18 Hz on 2017-era hardware. It feeds an eyes-free spatial-audio interface as a demonstration; we release the methodology as a reusable protocol for honest occupancy-completion claims.

1 Introduction

Any embodied agent acting in the physical world must reason about space it cannot directly observe — the occupancy occluded by the obstacles it moves among. A world model of only the visible surface is blind to half its workspace, and completing that hidden state is a prerequisite for reasoning about what happens next; for any downstream use, that inference is trustworthy only with an honest estimate of its own reliability: a confident hallucination of clear space behind a wall is worse than a calibrated "I don't know." Prior work addresses parts of this problem but not their conjunction (§5). **ShadowWeave**, a world model of occluded space, produces from a single monocular-depth frame (i) an egocentric BEV occupancy grid, (ii) an explicit shadow mask marking cells the sensor cannot see, and (iii) a calibrated per-cell occupancy probability inside that mask. Our contributions are:

- **A calibrated amodal-completion result.** From one depth frame, the model completes occluded occupancy with a micro-averaged shadow gain of +0.31–0.36 at 5 s (CIs clear of

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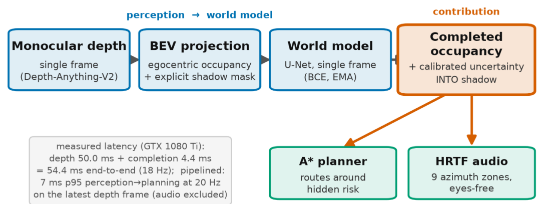

Figure 1: Pipeline: depth frame $\rightarrow$ 96×96 egocentric BEV occupancy and visibility; thresholded visibility gives the shadow mask; a 31M U-Net predicts occupancy at 1/3/5/10 s inside it, feeding A* planning and nine-zone HRTF audio.

zero at every horizon), and its in-shadow uncertainty is *better* calibrated (ECE 0.043–0.047 at 5 s) than in observed space.

- **A decomposition methodology** that separates completion from forecasting. It reveals that the gain is spatial completion of persistent hidden structure; the pure-forecasting residual is at or near zero (fixed static +0.000; moving tier +0.007 [+0.002, +0.012] at 5 s — significant but practically small, decaying with horizon as real forecasting should). We report this negative honestly: most pooled occupancy numbers in the literature share exactly this confound.
- **A memorization confound-kill.** Randomizing room geometry per episode leaves the gain statistically unchanged on never-seen rooms (+0.306 fixed vs +0.309 [+0.294, +0.324] randomized), showing completion is per-scene inference, not a memorized layout.
- **A real-time completion front-end** (7 ms p95 completion path, well inside 20 Hz; $\sim$18 Hz end-to-end with off-the-shelf depth on 2017-era hardware), feeding an A* + nine-zone HRTF eyes-free interface as a systems demonstration (§4).

2 Method

A differentiable projector (Fig. 1) lifts each monocular-depth image (rendered in simulation; from a monocular-depth network in deployment) into an egocentric 96×96 BEV occupancy grid covering 5 m ahead, plus a **visibility** grid: cells along a ray up to the first returned surface are observed-free, the surface cell observed-occupied, and everything beyond is **shadow**. Thresholding visibility at 0.5 yields the shadow mask — the region of interest for every metric here; optical flow between BEV frames is an optional channel. A 31M-parameter U-Net predicts occupancy from the single-frame BEV stack at 1/3/5/10 s as per-cell probabilities, trained with positive-up-weighted binary cross-entropy (BEV targets are $\approx$7% occupied) on 400 episodes with early stopping and EMA weights; we report the epoch-8 checkpoint (val. loss 0.300, IoU@5s 0.665). The input is a **single frame** with no multi-frame memory, making the completion-vs-forecasting distinction (§3) sharp. The per-cell probability *is* the uncertainty estimate; we test calibration **inside the shadow region specifically**, not merely grid-wide (where easy free space dominates). Downstream, the completed grid and its uncertainty drive an A* planner (extra cost through unobserved cells), a reactive collision-avoidance policy, and nine-zone HRTF audio cues; the audio interface is a demonstration, not user-evaluated.

3 Evaluation methodology

The empty-credit trap and micro-averaging. Our claim lives entirely inside a sparse, mostly-empty sub-region (shadow is $\sim$92% never-occupied), and the natural masked IoU scores an empty prediction against an empty masked region as a perfect 1.0; since most shadow cells are legitimately empty, this inflates both model and baselines, non-uniformly across sub-masks. We therefore report **micro-averaged** shadow IoU — pool integer intersection and union counts across all frames and divide once; empty denominators are undefined, never 1.0 — and the defended quantity is the **gain** over a persistence baseline (the observed frame carried forward), not the absolute level.

tier (fixed val, @5 s)	gain	95% CI
static	+0.306	[+0.288, +0.327]
moving	+0.363	[+0.349, +0.378]
debris	+0.315	[+0.281, +0.350]

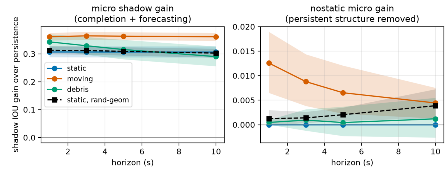

Figure 2: Micro shadow-IoU gain over persistence with 95% per-episode bootstrap CIs: per tier at 5 s, fixed geometry (left) and across horizons (right); every interval is clear of zero.

Completion vs. forecasting decomposition. All horizons are rendered from the same issue pose (static geometry projects to identical cells at every horizon), so each shadow cell is classified by its occupancy pattern across horizons: **STATIC** — occupied at *all* horizons (persistent hidden structure; ≈7% of shadow), where recall measures **amodal completion**; **DYNAMIC** — occupied at *some but not all* horizons (≤1.5%), where IoU measures genuine **forecasting** beyond persistence; **NEVER** — occupied at *no* horizon (≈92%), the false-positive test. (Dynamics faster than the 1 s grid — e.g. debris settling within a second — fall into STATIC, biasing the split toward completion; the continuously-moving tier is the cleaner forecasting test.) This separates completion from clairvoyance — a distinction pooled shadow-IoU hides.

Per-episode bootstrap CIs. Frames within an episode share geometry and pose, so we resample **episodes** with replacement ($B = 10{,}000$) and report the 2.5/97.5 percentile interval of the paired model-minus-persistence pooled micro-gain difference across replicates for every headline gain.

Calibration in shadow. We measure expected calibration error (ECE) restricted to shadow cells, contrasted with observed cells, using equal-mass binning (a fine 1000-bin histogram merged into 15 equal-count macro-bins, since in-shadow probabilities cluster near the prior and would leave equal-width bins empty).

4 Experiments

Setup. A MuJoCo single-room benchmark with three tiers — **static** (fixed obstacles), **moving** (kinematic movers), **debris** (objects that fall and settle) — 400 training / 80 validation episodes on disjoint seeds; per-tier analyses use 30-episode (9,000-frame) single-tier sets, randomized-geometry validation 80 episodes, and fixed-geometry results a 6×6 m room unless noted. Reproducibility is pinned: scene generation is bit-exact against a golden hash, and a rollout re-run reproduces every accuracy metric bit-identically (hardware-dependent latency excepted).

Shadow-gain headline. The model beats persistence at every horizon (Fig. 2), with every bootstrap interval clear of zero — statistically significant, not sampling noise. In policy rollouts the frame-averaged (macro) shadow gain is +0.325/+0.322/+0.320/+0.178 at 1/3/5/10 s. (Raw macro shadow-IoU levels — model 0.744 vs persistence 0.424 at 5 s — are empty-credit inflated and reported only for reference; the micro-gain is the defended number.)

Observation-blind prior. Persistence could be a weak baseline for *completion*, so we also evaluate a pose-marginal dataset prior (mean training-set occupancy in the egocentric frame, no observation at all), swept over thresholds and scored at its most favorable one. It reaches only 0.08–0.10 micro shadow IoU across tiers (0.09 on the randomized-geometry set): to recall hidden structure (0.86–0.92) it must blanket 72–79% of genuinely empty hidden space with false positives, whereas the model achieves 0.61–0.72 IoU at a 1.4–3.4% false-positive rate. An observation-*conditioned* heuristic (dilating observed obstacles into shadow, scored at its best radius) peaks at 0.31–0.35 — at or below persistence itself. Completion is thus learned observation-conditional inference; with geometry randomization (below), this rules out memorized layouts, marginals, and local extrapolation alike.

Completion, not clairvoyance. Decomposing the shadow mask attributes the gain almost entirely to amodal completion (Table 1). Excluding all persistent structure, the residual "nostatic" gain is at or near zero: fixed static +0.000, moving +0.007 (CI [+0.002, +0.012]), randomized-geometry static

Table 1: Top: shadow-mask decomposition at 5 s, fixed geometry (dynamic IoU spans 1 s→10 s): the gain is recall of persistent structure, not forecasting. Bottom: static-tier gain, fixed vs per-episode randomized geometry.

component (fixed val @5 s)	model	persistence
persistent-structure recall (amodal completion)	0.83–0.93	0.37–0.51
dynamic IoU, moving tier (1 s→10 s)	0.092 → 0.031	0.060 → 0.020
dynamic IoU, debris tier	0.020 → 0.019	0.006 → 0.023
false-positive rate on never-occupied cells	0.014–0.034	0.022–0.027

static tier, @5 s	micro shadow gain	95% CI	completion recall
fixed 6×6 geometry	+0.306	[+0.288, +0.327]	0.829
randomized geometry	**+0.309**	**[+0.294, +0.324]**	0.854

+0.002 ([+0.001, +0.004]) at 5 s. The honest claim is **amodal completion of hidden structure**, plus a small but statistically significant moving-tier forecasting signal that decays with horizon as genuine forecasting should. We do not claim dynamic forecasting on debris, where the dynamic signal is at the noise floor and persistence matches the model beyond 1 s.

Robustness to room geometry (memorization confound-kill). A fixed room shell could let the model memorize a constant wall prior, so we regenerated data with per-episode random width and depth (i.i.d. in $[5, 8]$ m, obstacle counts scaled to keep $\sim$7% positives), retrained from scratch, and evaluated on 80 held-out randomized rooms (Table 1, bottom): the gain does not drop (intervals overlap almost entirely) and completion recall is if anything higher on never-seen rooms: completion is per-scene inference, not a memorized shell.

Calibration in shadow. Shadow is $\sim$92% empty, so a raw in-shadow ECE is easy to make small; we therefore lead with the harder number — excluding persistent structure entirely, in-shadow ECE at 5 s remains a low 0.055–0.056 on the tiers with dynamic content (the static tier's structure-excluded region has no positive labels, hence no curve), so calibration is carried by the genuinely uncertain cells, not the easy always-occupied ones. On the full shadow mask at 5 s, in-shadow ECE is 0.043–0.047 — lower than observed space (0.058–0.086), Brier 0.022–0.036 — though this edge is aided by shadow's emptiness. Calibration is no fixed-room artifact: on randomized geometry, in-shadow ECE is 0.050–0.067 vs observed 0.056–0.077, shadow better-calibrated at 5–10 s (comparable at 1 s).

Ablations. Retraining without either auxiliary channel leaves the gain intact: no-visibility per-tier gains at 5 s (+0.309/+0.360/+0.313 vs full +0.306/+0.363/+0.315) are within CI and the no-flow rollout gain (+0.381 vs +0.320) is if anything higher; both are redundant (no optical flow on the critical path); as independent retrainings that all land within the full model's CIs, they also bound training-seed variance.

System. The perception-to-planning path runs at 6.47 ms p50 / 6.99 ms p95 (batch 1, deterministic U-Net), inside a 50 ms (20 Hz) budget; the excluded monocular-depth stage is not free (Depth-Anything-V2-Small: 49.9 ms p50 / 50.7 ms p95 in isolation, 54.4 ms measured end-to-end sequential, evaluation-era GTX 1080 Ti, 480×640), so deployment runs depth asynchronously while the 7 ms completion path consumes the latest depth frame at 20 Hz (audio synthesis excluded). An occupancy-arrival *diagnostic* (not falling-object anticipation; cell-level detection 0.787 at 2.62 s mean lead; component-level 0.752, precision only 0.714) counts events per due prediction, not per object, keys on cells revealed rather than newly occupied (occluded static geometry revealed by agent motion also counts), and its lead is bounded by the $\{1,3,5,10\}$ s horizon grid, missing the $\geq$3 s target; we report it as a diagnostic, not a strength. Navigation: per-episode collision rate 0.367 (vs 0.60 untrained; straightness 0.773), missing the $\leq$0.10 target — an honest demonstration, no more.

Generative variant. A conditional-diffusion (DDPM) world model is ongoing work: iterated DDIM sampling exceeds our batch-eval budget; a like-for-like comparison is future work.

5 Related work

Predicting the structure of unobserved space has been approached from three angles. Semantic scene completion, from SSCNet [Song et al., 2017] through MonoScene [Cao and de Charette, 2022] and its monocular successors VoxFormer [Li et al., 2023] and VisHall3D [Lu et al., 2025], infers dense voxel geometry — and even hallucinates geometry beyond the visible frontier — from a single image, but targets outdoor driving voxels evaluated purely by mIoU/IoU, without calibrated uncertainty in the occluded region. Occupancy world models such as OccWorld [Zheng et al., 2024], together with occupancy-forecasting benchmarks (Occ3D; Tian et al., 2023) and self-supervised raycasting forecasters [Khurana et al., 2023], instead predict the *temporal* evolution of multi-frame occupancy for autonomous vehicles. Closest in spirit, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single view for navigation, yet emits only point estimates. ShadowWeave is distinct: egocentric BEV amodal completion into explicitly masked occluded space from one monocular-depth frame, with calibrated uncertainty inside the hidden region.

6 Conclusion and limitations

Calibrated amodal completion of occluded occupancy works from one monocular-depth frame, transfers to never-seen rooms, and fits a real-time perception-to-planning budget; the empty-credit-immune micro-gain, per-episode CIs, and completion/forecasting decomposition keep the claim honest and, we hope, get reused. The limitations are those it surfaces: evaluation is in simulation on clean simulator depth (robustness to estimated-depth noise is untested), so we make no quantitative real-world claim, though the front-end already accepts real monocular depth (Depth-Anything-V2), making sim-to-real transfer the natural next step; baselines are self-constructed (prior methods target different input regimes); the model performs amodal *completion*, not multi-step *forecasting*, and is framed accordingly; navigation misses its collision target and the audio interface is not user-evaluated (system context); and falling-object lead is bounded by the horizon grid.

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